

CS 352

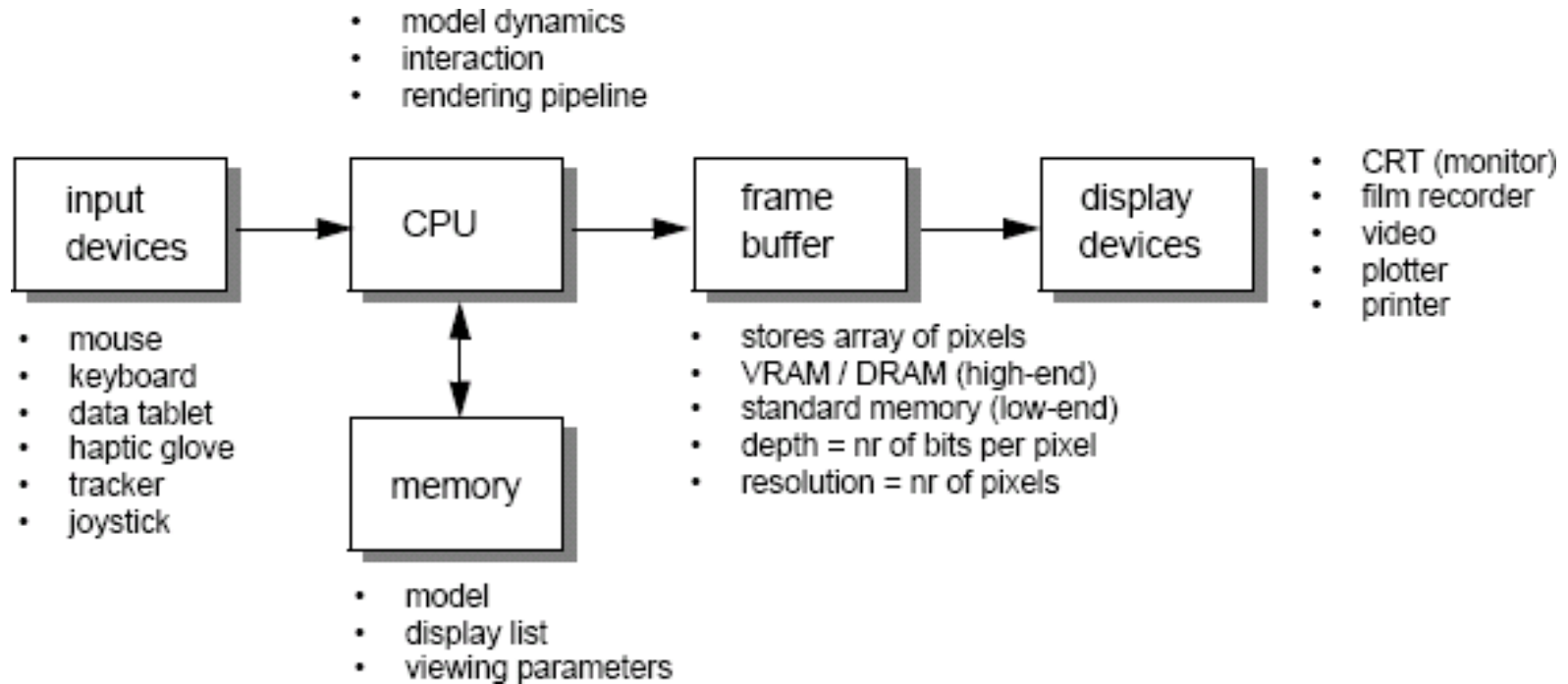
Computer Graphics

Lecture 3

Overview of graphics

Hardware Devices

Overview of a graphics system



Frame Buffer

- The frame buffer contains the image stored in binary form as a matrix of 0's and 1's which represent the pixel. 0 indicates the darkness and 1 indicates the image.
- The **Frame Buffer** holds the set of intensity values for all the **screen points**.
- The intensity values stored in a Frame Buffer are retrieved and painted on a screen one row at a time. This row is called as **scan line**.

Graphics Input Device

- ❑ The graphics input devices are devices that allow the information from outside the computer to be communicated with the computer.
- ❑ The graphics input devices are divided mainly into 2 groups:

➤ Analog devices such as:

- ✓ Track ball
 - ✓ Joystick
 - ✓ Mouse
 - ✓ Image Scanner.
- The analog input devices need analog to digital converter (A/D) to change the analog signals to a digital form understandable by the processors.



• Digital devices such as:

- ✓ Keyboard ,
- ✓ Digitized camera
- ✓ Data glove: a glove with sensors embedded at finger joints to measure the bending of the fingers. Used to control a virtual hand for grasping, dropping, and moving an object in a virtual environment.
- ✓ Touch panel: highly transparent and embedded over a display surface.
- ✓ Light Pen: When the beam is light touched the output screen and integral button pressed, the processor is able to detect the fact and ascertain location of the position pointed to by the light beam.

Graphics Input Device

Feature	Mouse	Keyboard	Joystick	Trackball
Input method	Moving the mouse	Typing	Moving the joystick	Rotating the trackball
Typical uses	General computer use, selecting items, moving the cursor, and dragging objects	Entering text, commands, and other information into a computer	Playing video games, controlling robots, and other applications that require precise control of objects	General computer use, especially for laptops and other devices with limited space
Advantages	Precise cursor control, easy to use, affordable	Fast and efficient for text input, many keyboard shortcuts available	Immersive gaming experience, good for precise control of objects	Portable, easy to use, less strain on the wrist
Disadvantages	Can be tiring to use for long periods of time, can cause repetitive strain injury	Not suitable for gaming or other applications that require precise control of objects	Can be difficult to use for text input, not as portable as other input devices	Can be difficult to use for precise control of objects, not as portable as other input devices

Graphics Output Devices

□ The graphical output devices can be categorized as:

- **Hardcopy Graphics Devices:** These are devices that produce physical, permanent output, such as printed documents or images. Examples include printers, plotters, and 3D printers.
- **Softcopy Graphics Devices:** These are devices that display temporary, non-permanent output on a screen. Examples include monitors, projectors, and virtual reality (VR) headsets.

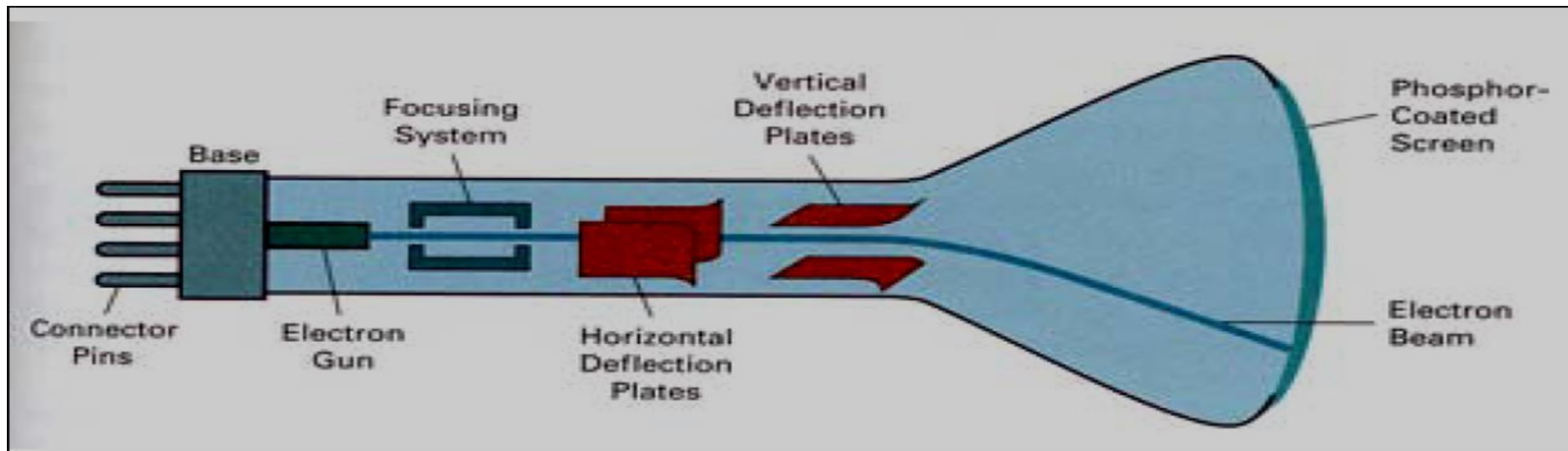
These devices have a capability of producing a graphical outputs, some common of these graphical devices are:

1. Raster Scan Cathode Ray Tube (CRT).
2. Plasma Display (like the one used in portable case computers).
3. Liquid Crystal Display (LCD) (like one used in games, watches, ...).
4. Plotters and Printers.

comparison between hardcopy and softcopy graphics devices based on their characteristics, applications, advantages, and disadvantages

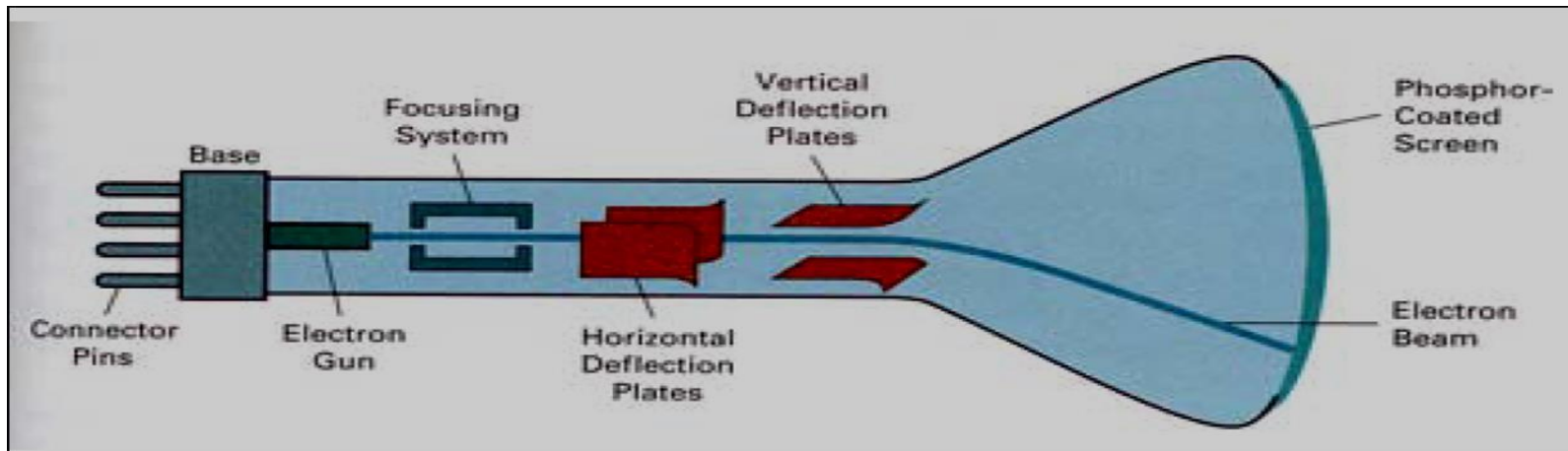
Aspect	Hardcopy Devices	Softcopy Devices
Output Type	Physical, permanent	Digital, temporary
Examples	Printers, plotters, 3D printers	Monitors, projectors, VR headsets
Portability	Portable (e.g., printed documents)	Requires a device to display content
Interactivity	Static output	Dynamic and interactive
Cost	High (consumables and maintenance)	Lower (no consumables, but hardware costs)
Environmental Impact	Higher (paper waste, ink/toner usage)	Lower (eco-friendly, no physical waste)
Editing Flexibility	Limited (cannot modify after printing)	High (easy to edit and update)

Cathode Ray Tube (CRT)



- **Electron gun** is the primary component of a CRT. When the heat is supplied to the electron gun by directing a current, a **beam of electrons** emitted by an electron gun, passes through focusing and deflection systems that direct the beam toward specified positions on the phosphor-coated screen.
- **The focusing system** in a CRT is needed to force the electron beam to converge into a small spot as it strikes the phosphor.
- There are two pairs of **deflection plates** - **Horizontal deflection plates** and **vertical deflection plates**.
- One pair of plates is mounted horizontally to control the vertical deflection, and the other pair is mounted vertically to control horizontal deflection.

Cathode Ray Tube (CRT)



- The beam passes between the two pairs of deflection plates and is positioned on the screen.
- The **phosphor** then emits a small spot of light at each position contacted by the electron beam.
- Because the light emitted by the phosphor fades very rapidly, some method is needed for maintaining the screen picture.
- One Way to keep the phosphor glowing is to redraw the picture repeatedly by quickly directing the electron beam back over the same points. This type of display is called a **refresh CRT**.
- In CRT monitors there are *two techniques of displaying images:*
 - 1) **Raster scan displays**
 - 2) **Random scan displays**

Raster Scan Displays

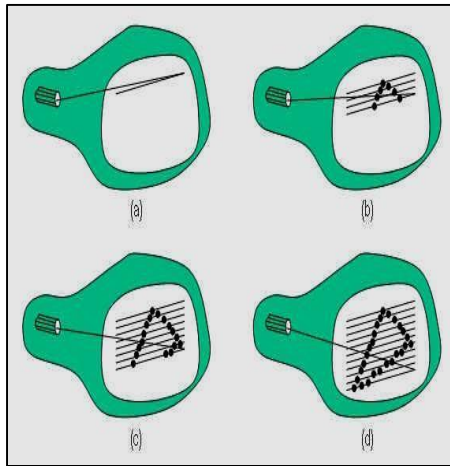


Figure 5: Raster Scan Display

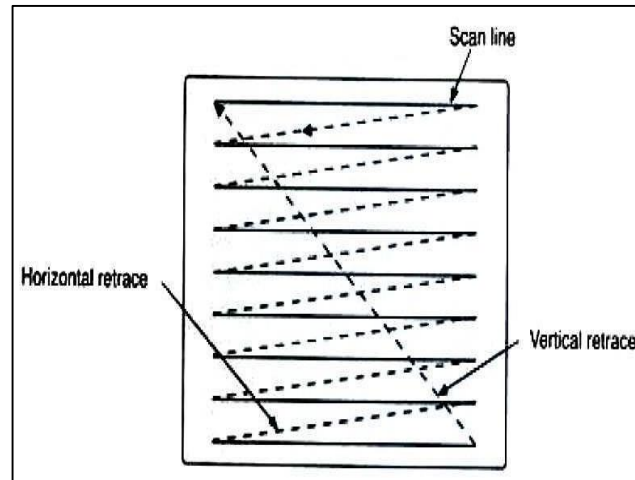


Figure 6: Horizontal and Vertical Retrace

- ❑ The most common type of graphics monitor employing a CRT is the raster-scan display, based on television technology.
- ❑ In a raster-scan system, the electron beam is swept across the screen, one row at a time, from top to bottom. Each row is referred to as a scan line.

Retracing:

At the end of each scan line, the electron beam returns to the left side of the screen to begin displaying the next scan line.

Horizontal retrace

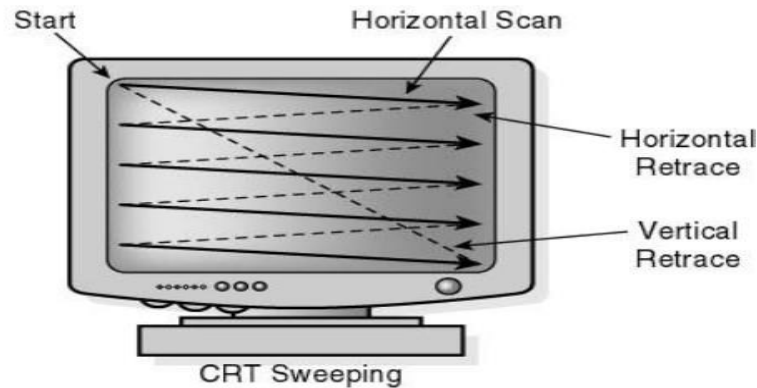
The return to the left of the screen, after refreshing each scan line.

Vertical retrace

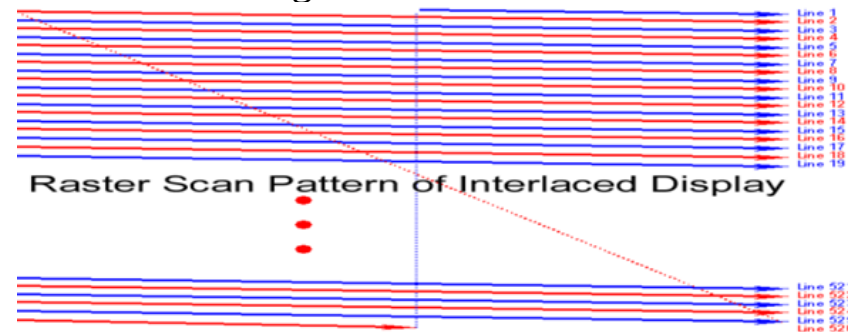
At the end of each frame, the electron beam returns to the top left corner of the screen to begin the next frame.

Raster-Scan Displays

- The return to the left of the screen, after refreshing each scan line is called **Horizontal retrace**.
- At the end of each frame the electron beam returns to the top left corner of the screen to begin the next frame is called **Vertical retrace**.
- **To reduce flicker**, divide frame into two fields—one consisting of the even scan lines and the other of the odd scan lines.
- Even and odd fields are scanned out alternately to produce an interlaced image.

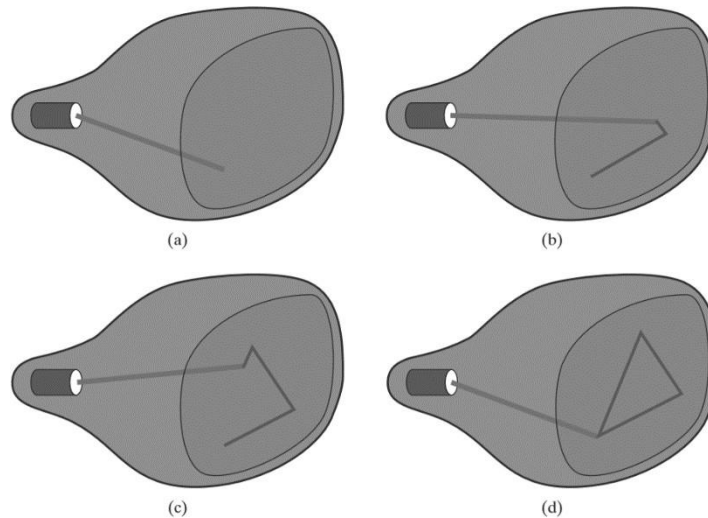


- **Interlacing** is a technique used in video displays to improve the perceived motion and reduce flickering. It involves dividing a video frame into two interlaced fields, each containing half of the horizontal lines of the frame. The first field consists of all the odd-numbered lines, while the second field contains all the even-numbered lines.
- **Interlacing**: the refresh cycle on raster scan monitors is carried out by sweeping the beam over the even frame from top to the bottom and repeating it for the odd frame (i.e. the raster cycle is broken into 2 sweep pattern. Each of it consisting of half the number of raster lines in the original scan pattern).



Random-Scan Displays

- CRT has the electron beam directed only to those parts of the screen where a picture is to be displayed. Pictures are generated as line drawings, with the electron beam tracing out the component lines one after the other. For this reason, random-scan monitors are also referred to as **vector displays** (or stroke-writing displays or calligraphic displays).



Computer graphics displays techniques

Feature	Raster Display	Random Display
Image Formation	Scans the entire screen line by line, turning pixels on or off	Draws lines or shapes directly on the screen
Resolution	Generally lower	Generally higher
Cost	More affordable	More expensive
Modification	Difficult to modify individual elements	Easier to modify individual elements
Interlacing	Commonly used to reduce flicker	Not used
Rendering	Uses pixels to create imagery	Uses mathematical functions to create lines and shapes
Screen Refresh	Entire screen refreshed at a constant rate (e.g., 60Hz)	Only displayed area refreshed, refresh rate depends on complexity
Suitable for	Realistic images, photographs, videos	Line drawings, vector graphics, scientific visualizations
Examples	TVs, computer monitors, smartphones	Oscilloscopes, early computer monitors (rare in modern times)

Computer graphics displays techniques

RASTER vs. VECTOR

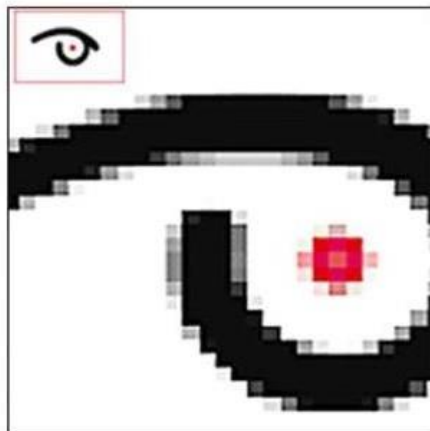
When using graphics programs to set-up your files, it is extremely important to understand the difference between a raster file and a vector file.

RASTER

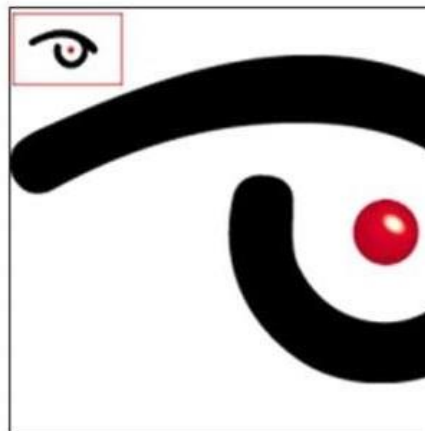
- Defined by pixels
- Non-scaleable
- Resolution dependent
- Continuous tone, photographic qualities
- Always use for photos

VECTOR

- Defined by mathematical equations (lines & curves)
- Scaleable to any size
- Independent of resolution
- Cartoon-like, line drawing qualities
- Always use for logos, text



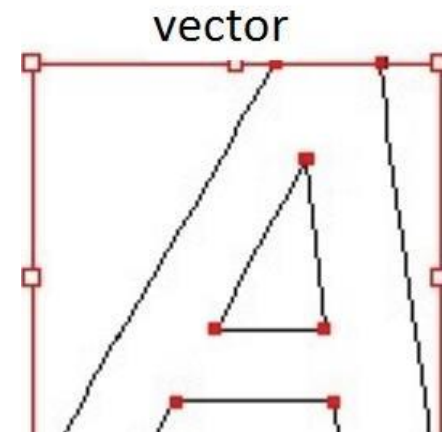
RASTER FILE ENLARGED



VECTOR FILE ENLARGED



raster



vector

Image quality Vs. Image Resolution

Feature	Image quality	Image resolution
Definition	How good an image looks, including its sharpness, contrast, color accuracy, and noise level.	The number of pixels per unit of length in an image.
Factors that affect	Compression, sensor size, lens quality, lighting, and scene complexity.	Image size, pixel density, and display size.
Impact on viewing experience	Higher image quality results in a more realistic and immersive viewing experience.	Higher image resolution results in a sharper and more detailed image, especially when viewed on a large display.
Examples	A high-quality photo with accurate colors and sharp details, even when viewed closely.	A high-resolution image that is large enough to be printed at a large size without losing detail.

Image quality Vs. Image Resolution

Relationship between image quality and image resolution

❑ Image quality and image resolution are related, but they are not the same thing.

- ✓ An image can have a high resolution but low quality, and vice versa. For example, an image that has been compressed too much may have a high resolution but low quality. The compression process can discard image data, which can lead to artifacts such as blurring and pixelation.
- ✓ On the other hand, an image with a low resolution can have high quality if it has been captured with a good sensor and lens and under good lighting conditions.

❑ Which is more important?

- ✓ If the image is going to be viewed on a small display, such as a smartphone screen, then image quality is more important than image resolution. A high-quality image will look good even at a small size.
- ✓ However, if the image is going to be printed at a large size, then image resolution is more important. A high-resolution image will ensure that the printed image is sharp and detailed.

The Factors that affect the Image quality

1. Resolution

- **Definition:** The number of pixels in an image, typically expressed as width × height (e.g., 1920 × 1080).
- **Impact:** Higher resolution results in sharper and more detailed images. Low resolution can lead to pixelation and loss of detail.
- **Examples:**
 - **Cameras:** Higher megapixel cameras produce higher-resolution images.
 - **Displays:** Monitors with higher resolutions (e.g., 4K) display images more clearly.

2. Bit Depth

- **Definition:** The number of bits used to represent the color of each pixel.
- **Impact:** Higher bit depth allows for more colors and smoother gradients, reducing banding and improving color accuracy.
- **Examples:**
 - **8-bit:** 256 colors per channel (common in standard displays).
 - **10-bit or 12-bit:** Thousands to millions of colors (used in professional photography and high-end displays).

3. Compression

- **Definition:** The process of reducing file size by removing redundant or less important data.
- **Impact:** Excessive compression (e.g., JPEG) can lead to artifacts, loss of detail, and reduced image quality. Lossless compression (e.g., PNG) preserves quality but results in larger file sizes.
- **Examples:**
 - **JPEG:** Lossy compression, commonly used for web images.

PNG: Lossless compression, used for high-quality images

The Factors that affect the Image quality

4. Lighting Conditions

- **Definition:** The amount and quality of light when capturing an image.
- **Impact:** Poor lighting (e.g., low light or harsh shadows) can result in noise, underexposure, or overexposure. Good lighting enhances clarity, color accuracy, and detail.
- **Examples:**
 - **Natural Light:** Produces balanced and vibrant images.

Artificial Light: Can cause color casts or uneven lighting.

5. Lens Quality

- **Definition:** The optical quality of the lens used to capture an image.
- **Impact:** High-quality lenses produce sharper images with minimal distortion, chromatic aberration, and vignetting. Poor-quality lenses can result in blurry or distorted images.
- **Examples:**
 - **Prime Lenses:** Fixed focal length, often sharper and faster.
 - **Zoom Lenses:** Variable focal length, may sacrifice some sharpness.

6. Noise

- **Definition:** Grain or speckles in an image, often caused by low light or high ISO settings.
- **Impact:** High noise reduces image clarity and detail. Noise reduction techniques can help but may soften the image.
- **Examples:**
 - **Low ISO:** Produces cleaner images with less noise.
 - **High ISO:** Increases sensitivity but introduces noise.

The Factors that affect the Image quality

7. Color Accuracy

- **Definition:** The faithfulness of colors in an image to the original scene.
- **Impact:** Poor color accuracy can result in unnatural or washed-out colors. Proper white balance and color calibration are essential.
- **Examples:**
 - **White Balance:** Adjusts colors to match the lighting conditions.
 - **Color Profiles:** Ensure consistent color reproduction across devices.

8. File Format

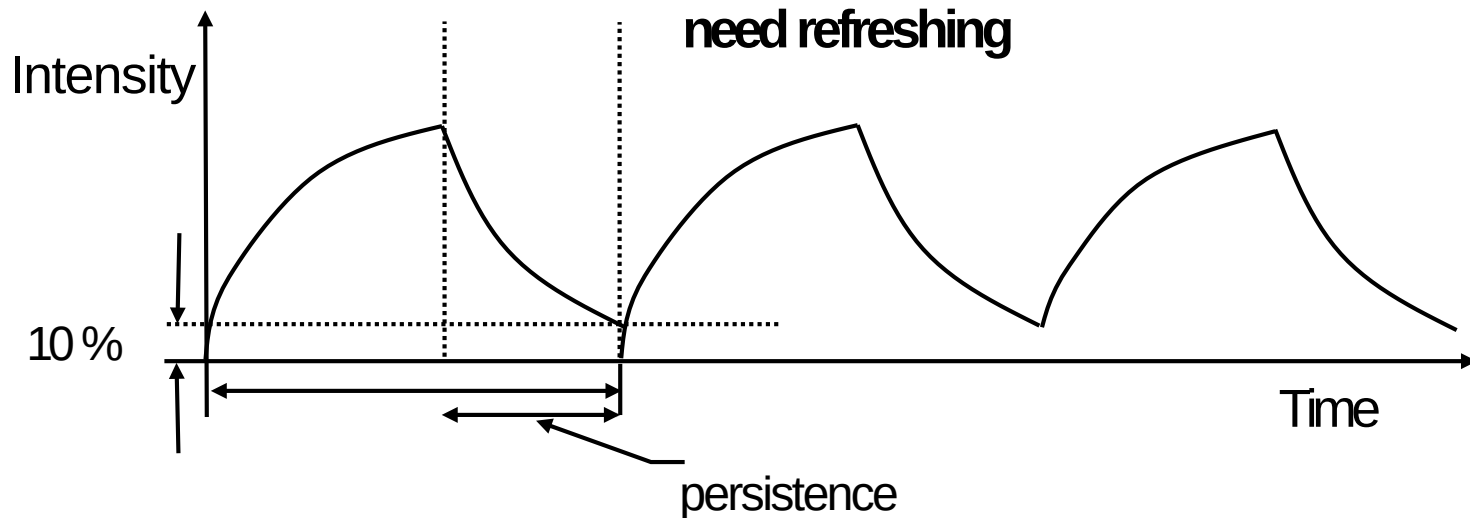
- **Definition:** The format in which an image is saved (e.g., JPEG, PNG, RAW).
- **Impact:** Different formats have varying levels of compression and quality preservation.
- **Examples:**
 - **RAW:** Uncompressed, retains maximum quality for editing.

JPEG: Compressed, smaller file size but lower quality

9. Environmental Factors

- **Definition:** External conditions during image capture or display.
- **Impact:** Dust, humidity, and temperature can affect both capture and display quality.
- **Examples:**
 - **Dust on Lens:** Reduces sharpness and clarity.
 - **Humidity:** Can damage equipment or affect display performance.

Persistence



- **Persistence:** How long a phosphor continues to emit light after the electron beam is removed
- The **persistence** of a phosphor is defined as the **time** it takes for emitted light to decay to **1/10** of its original intensity

Refresh rate

- # of times/sec a picture is *redrawn*
required to be *flicker-free*
- 60 Hz for raster scan displays
- For refresh displays, it depends on picture complexity !!!

Relationship between refresh rate and phosphor persistence

- ❑ **Refresh rate** is a measure of how often the image on a display is updated, while phosphor persistence is a measure of how long it takes for a phosphor dot to decay to half of its original brightness.
 - ✓ A higher refresh rate can help to reduce flicker and make images appear smoother and more fluid. However, if the phosphor persistence is too high, then the image may still appear blurry or ghosted, even at a high refresh rate.
 - ✓ A lower phosphor persistence can help to reduce motion blur and ghosting. However, if the refresh rate is too low, then the image may appear to flicker.
- ❑ For most users, a refresh rate of 60 Hz or higher is sufficient to eliminate flicker. However, users who play fast-paced games or watch a lot of fast-moving video content may prefer a higher refresh rate, such as 120 Hz or 144 Hz.
- ❑ **Phosphor persistence** should be as low as possible to reduce motion blur and ghosting. However, it is important to note that some displays, such as plasma displays, have a naturally higher phosphor persistence than others.
- ❑ Both refresh rate and phosphor persistence are important factors to consider when choosing a display. However, refresh rate is generally considered to be more important, as it has a greater impact on the overall fluidity of the image.

Relationship between refresh rate and phosphor persistence

Feature	Refresh rate	Phosphor persistence
Definition	The number of times per second that the image on a display is refreshed.	The amount of time it takes for a phosphor dot to decay to half of its original brightness after being excited by an electron beam.
Unit	Hertz (Hz)	Milliseconds (ms)
Impact on viewing experience	Higher refresh rates reduce flicker and make images appear smoother and more fluid.	Higher phosphor persistence can lead to motion blur and ghosting, while lower phosphor persistence can make images appear flicker-prone.
Ideal values	For most users, a refresh rate of 60 Hz or higher is sufficient to eliminate flicker. Phosphor persistence should be as low as possible to reduce motion blur and ghosting.	
Examples	❑ A CRT monitor with a refresh rate of 60 Hz and a phosphor persistence of 25 ms. ❑ An LCD monitor with a refresh rate of 144 Hz and a phosphor persistence of 1 ms.	

Examples

Problem 1: Refresh Rate

A computer monitor has a refresh rate of 60 Hz. Calculate the time it takes for one complete refresh cycle.

Solution:

- The refresh rate is given as 60 Hz, which means the monitor refreshes the display 60 times per second.
- The time for one complete refresh cycle:
- $\text{Time} = 1 / \text{Refresh Rate}$
- $\text{Time} = 1 / 60 \text{ Hz}$
- $\text{Time} = 0.0167$ seconds or approximately 16.7 milliseconds
- Therefore, it takes approximately 16.7 milliseconds for one complete refresh cycle on this monitor.

Examples

Problem 2: Phosphor Persistence

A cathode-ray tube (CRT) display has a **phosphor persistence of 10 milliseconds**. Calculate the *minimum refresh rate required to avoid visible flickering on the screen*.

Solution:

- The phosphor persistence is given as 10 milliseconds, which means that the phosphors on the CRT screen take 10 milliseconds to fade away after being excited.
- To avoid visible flickering, the refresh rate should be higher than the phosphor persistence.
- Let's assume the desired minimum refresh rate *is R Hz*.
- Therefore, the time for one complete refresh cycle should be less than or equal to the phosphor persistence:
- Time for one complete refresh cycle $\leq$ Phosphor Persistence
- $1 / R \leq 10$ milliseconds
Hz $\implies R \geq 1 / 10$ milliseconds $\implies R \geq 100$
- **Therefore, the minimum refresh rate required to avoid visible flickering on the screen should be at least 100 Hz.**

Examples

problem 3:

A CRT monitor has a refresh rate of 60 Hz and a phosphor persistence of 25 ms.
What is the percentage of time that the monitor is displaying a blank screen?

Solution:

- ❑ The monitor has a refresh rate of 60 Hz and phosphor persistence of 25 ms.
- ❑ **The percentage of time that the monitor is displaying a blank screen is equal to the phosphor persistence divided by the refresh rate.**

The percentage of time that the monitor is displaying a blank screen is :

$$25 \text{ ms} / 60 \text{ Hz} = 0.417\%.$$

Examples

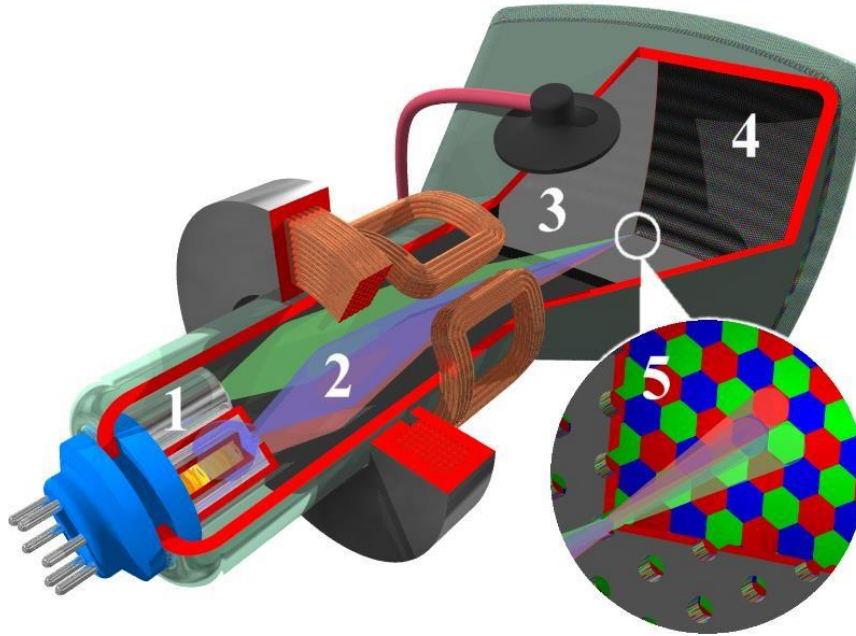
problem 4:

A gamer is playing a fast-paced game on a CRT monitor. The game requires the monitor to display images with minimal motion blur. The monitor has a refresh rate of 60 Hz and a phosphor persistence of 25 ms. Is the monitor's refresh rate fast enough to eliminate motion blur?

Solution:

- ❑ Motion blur is caused by the persistence of the phosphor dots on a CRT monitor. When the monitor displays a moving image, the phosphor dots remain lit for a short period of time after they are excited by the electron beam. This causes the image to appear blurry when it is moving quickly.
- ❑ To eliminate motion blur, the monitor's refresh rate must be fast enough to ensure that each phosphor dot has decayed to a sufficiently low brightness before the next image is displayed. In this case, the monitor has a refresh rate of 60 Hz and a phosphor persistence of 25 ms. This means that each phosphor dot has 16.67 ms ($1000 \text{ ms} / 60 \text{ Hz}$) to decay to a sufficiently low brightness before the next image is displayed.
- ❑ Unfortunately, 16.67 ms is not enough time for the phosphor dots to decay to a sufficiently low brightness. This means that the monitor's refresh rate is not fast enough to eliminate motion blur.

Color Cathode Ray Tube (cont.)

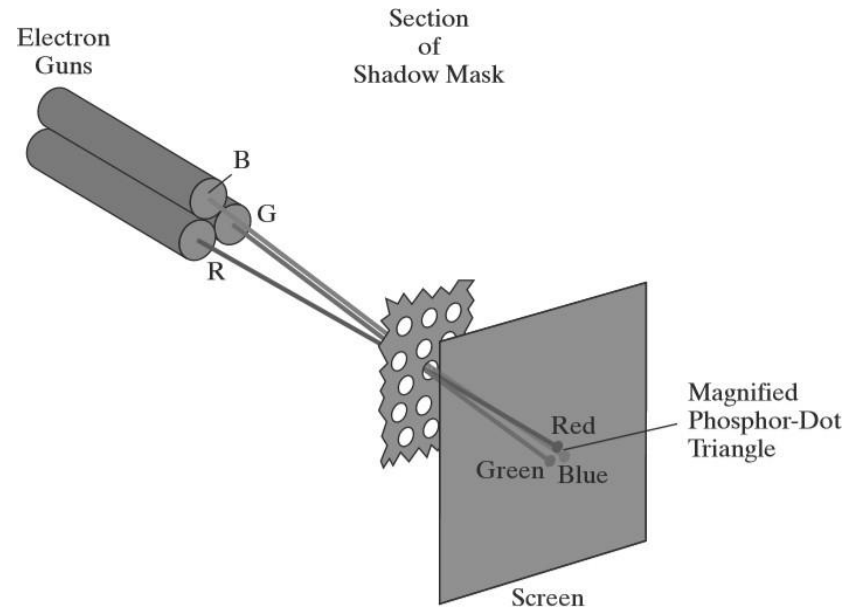


The interior of a cathode-ray tube for color televisions and monitors:

- ١. Electron guns
- ٢. Electron beams
- ٣. Mask for separating beams for red, green and blue part of displayed image
- ٤. Phosphor layer with red, green, and blue zones
- ٥. Close-up of the phosphor-coated inner side of the screen

Color CRT Monitors

- This type of CRT has **three electron guns**, one for each color dot, and a shadow-mask grid just behind the phosphor-coated screen.
- When the three beams pass through a hole in the shadow mask, they activate a dot triangle, which appears as a small color spot on the screen
- We obtain color variations in a shadow-mask CRT by varying the intensity levels of the three electron beams.



Methods of producing color on a cathode ray tube (CRT) display.

1) Beam-penetration:

- ❑ In the beam penetration method, there is a single electron gun that fires a beam of electrons at a multi-layered phosphor screen. The different layers of phosphor emit different colors of light when they are struck by the electron beam. The color of the pixel is determined by which layer of phosphor is struck by the electron beam.
- ❑ Two layers of phosphor (red ,green) are coated inside CRT screen.
- ❑ The displayed color depends on how far the electron beam penetrate phosphor layers.
- ❑ It has only 4 colors
- ❑ The beam penetration method is a simpler and less expensive way to produce color CRT monitors, but it has several disadvantages. First, it has a limited color range, as it can only produce a few different colors. Second, the picture quality is poor, as the electron beam can excite phosphor dots of the wrong color. Third, the resolution is lower than that of shadow mask monitors.

Methods of producing color on a cathode ray tube (CRT) display.

2) Shadow-mask :

- ❑ In the shadow mask method, there are three electron guns, one for each color (red, green, and blue). The electron guns fire beams of electrons at a phosphor screen that is covered by a shadow mask.
- ❑ The shadow mask contains a set of holes aligned with the phosphor pattern, when the beams pass through a hole in the shadow mask, it activate a dot triangle as in figure.
- ❑ The shadow mask method is a more complex and expensive way to produce color CRT monitors, but it has several advantages over the beam penetration method. First, it has a wider color range, as it can produce a wide variety of colors by varying the intensity of the electron beams. Second, the picture quality is good, as the electron beam is more focused and less likely to excite the wrong phosphor dots. Third, the resolution is higher than that of beam penetration monitors.
- ❑ Beam penetration and shadow mask methods are now obsolete technologies. They have been replaced by modern monitors, such as LCD, LED, and OLED monitors. Modern monitors offer a better combination of features than CRT monitors. They have a wider color range, viewing angle, and refresh rate than CRT monitors. They are also lighter, more energy-efficient, and more affordable than CRT monitors.

Beam Penetration Vs. Shadow mask

Feature	Beam Penetration	Shadow Mask
Technology	Utilizes a single electron beam	Utilizes multiple electron beams and a shadow mask
Color	Monochrome (single color)	Color (multiple colors)
Resolution	Lower resolution	Higher resolution
Image Quality	Lower image quality	Higher image quality
Dot Pitch	Larger dot pitch (less sharp image)	Smaller dot pitch (sharper image)
Manufacturing	Simpler manufacturing process	More complex manufacturing process
Cost	Lower cost	Higher cost
Application	Older CRT displays	CRT displays with color capabilities
Color Accuracy	Limited color accuracy	Improved color accuracy
Refresh Rate	Generally lower refresh rates	Higher refresh rates
Screen Size	Suitable for smaller screen sizes	Suitable for larger screen sizes

Flat-Panel Displays

- The term flat-panel display refers to a class of video devices that have reduced volume, weight, and power requirements compared to a CRT.
- We can separate flat-panel displays into two categories: **emissive** displays and **non-emissive** displays.
- The **emissive** displays (or emitters) are devices that convert electrical energy into light such as Plasma panels and light-emitting diodes LED.
- **Non-emissive** displays (or nonemitters) use optical effects to convert sunlight or light from some other source into graphics patterns. The most important example of a non-emissive flat-panel display is a liquid-crystal device LCD.

Flat-Panel Displays

light-emitting diode (LED)

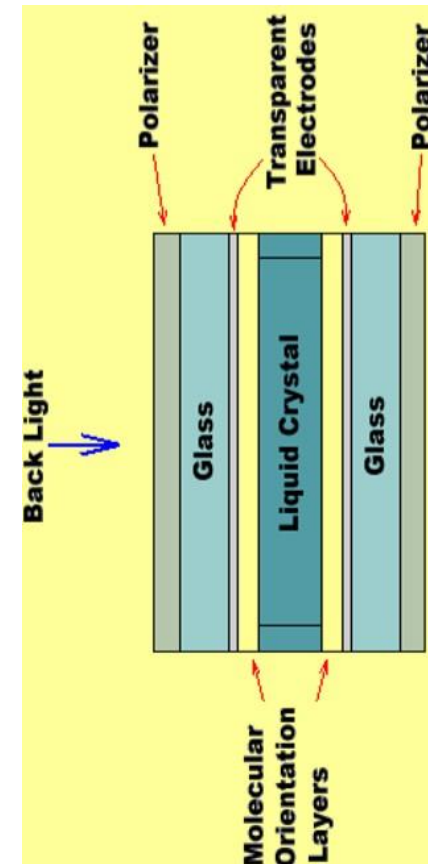
- A matrix of diodes is arranged to form the pixel positions in the display, and picture definition is stored in a refresh buffer. As in scan-line refreshing of a CRT, information is read from the refresh buffer and converted to voltage levels that are applied to the diodes to produce the light patterns in the display.

Liquid-crystal displays (LCDs)

- Are commonly used in small systems, such as laptop computers and calculators. These non-emissive devices produce a picture by passing polarized light from the surroundings or from an internal light source through a liquid-crystal material that can be aligned to either block or transmit the light.

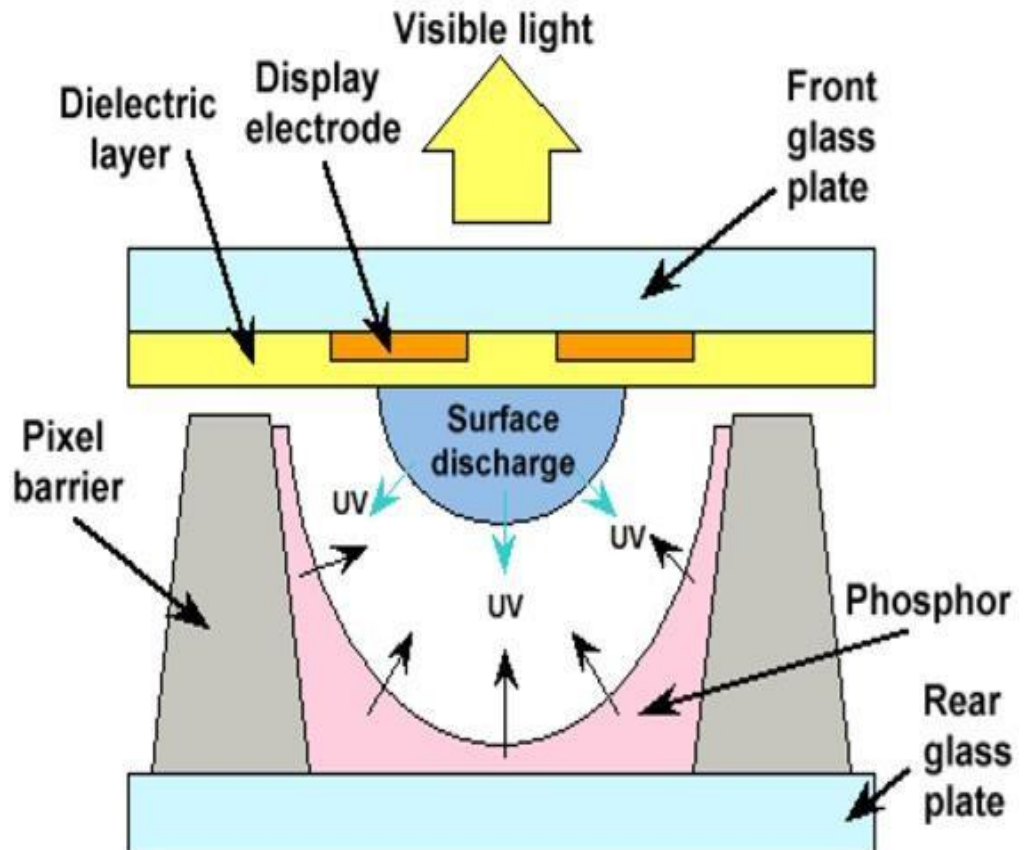
Liquid Crystal Displays (LCDs)

- A liquid-crystal display (LCD) is a flat-panel display or other electronically modulated optical device that uses the light-modulating properties of liquid crystals combined with polarizers.
- Liquid crystals do not emit light directly, instead using a backlight or reflector to produce images in color or monochrome.
- LCDs are available to display arbitrary images (as in a general-purpose computer display) or fixed images with low information content, which can be displayed or hidden.



Plasma Display Panels

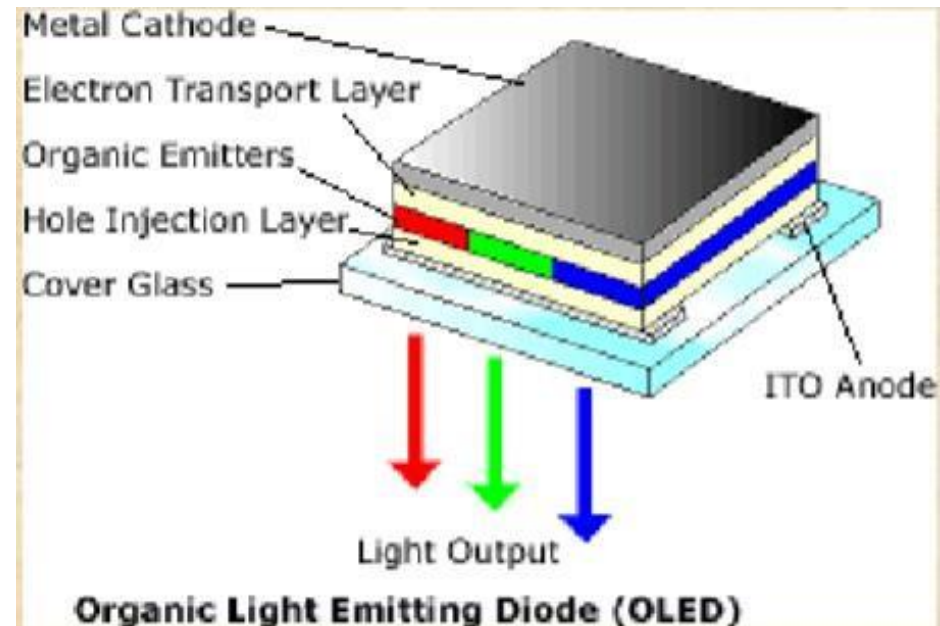
- Promising for large format displays
- Basically fluorescent tubes
- High-voltage discharge excites gas mixture (He, Xe)
- Upon relaxation UV light is emitted
- UV light excites phosphors
- Large viewing angle
- Less efficient than CRTs
- Not as bright
- More power
- No refresh required



	LCD TV	Plasma TV
Thickness	Minimum 1 inch	Minimum 1.2 inches
Power consumption	Requires less power to operate <u>when compared to plasma, but more</u> than OLED TVs	Consumes slightly more power than an LCD TV. Modern plasma displays receive high Energy Star (US) ratings.
Screen size	13 - 57 inches	42 inches and above
Cost	Much cheaper	Cheaper than <u>LED-lit TVs</u>
Life span	50,000 - 100, 000 hours	Around 20, 000 – 60,000 hours

Light Emitting Diode (LED)

- Organic Light Emitting Diodes (OLEDs)
- Function is similar to a semiconductor LED
- Thin-film polymer construction
- Potentially simpler processing
- Transparent
- Flexible
- Can be vertically stacked
- Excellent brightness



Basis For Comparison	LED	LCD
Definition	PN-Junction device which discharge visible lights when an electrical charge passes through it.	It is an optical device used for displaying the information in the form of text and images.
Stand For	Light Emitting Diode	Liquid Crystal Display
Resolution	High	Low
Power Requirement	More	Less
Display Area	Small	Large
Cost	High	Low

PARAMETER	LED	OLED
Expands to	Light emitting diode	Organic light emitting diode
Power consumption	More	Less
Cost	Less	More
Picture quality	Good	Comparatively better
Brightness	More	Less
Size	Large	Comparatively small
Lifespan	More	Less
Screen Thickness	Thin	Comparatively thinner
Weight	Heavy	Lighter in comparison to LED
Response time	Slow	Fast

Different technologies used to manufacture touch panels.

It is manufactured using different technologies (Resistive - Capacitance - Infrared).

- ❑ **Resistive**→ A chip inside the screen figures out the coordinates of the place you touched. When you press a resistive touchscreen, you push two conducting layers together so they make contact, a bit like an ordinary computer keyboard. These screens are made from multiple layers of glass.
- ❑ **Capacitance**-> When a capacitive panel is touched, it draws a small charge to the point of contact, becoming a functional capacitor. The change in the electrostatic field is measured to find the location. In some designs, circuits located at each corner of the panel calculate the charge and send the information to the controller for processing.
- ❑ **Infrared**-> The infrared touchscreen works by detecting interruptions of infrared beams emitted by LEDs embedded in the frame around the touchscreen overlay. There are two rows of IR LEDs, which generate horizontal and vertical invisible IR beams. They form a large beam grid that covers the surface of the overlay.